Questions

Q1.

Balloon **P** hangs from an insulating thread.

A teacher gives the balloon a positive electric charge, as shown in Figure 15.

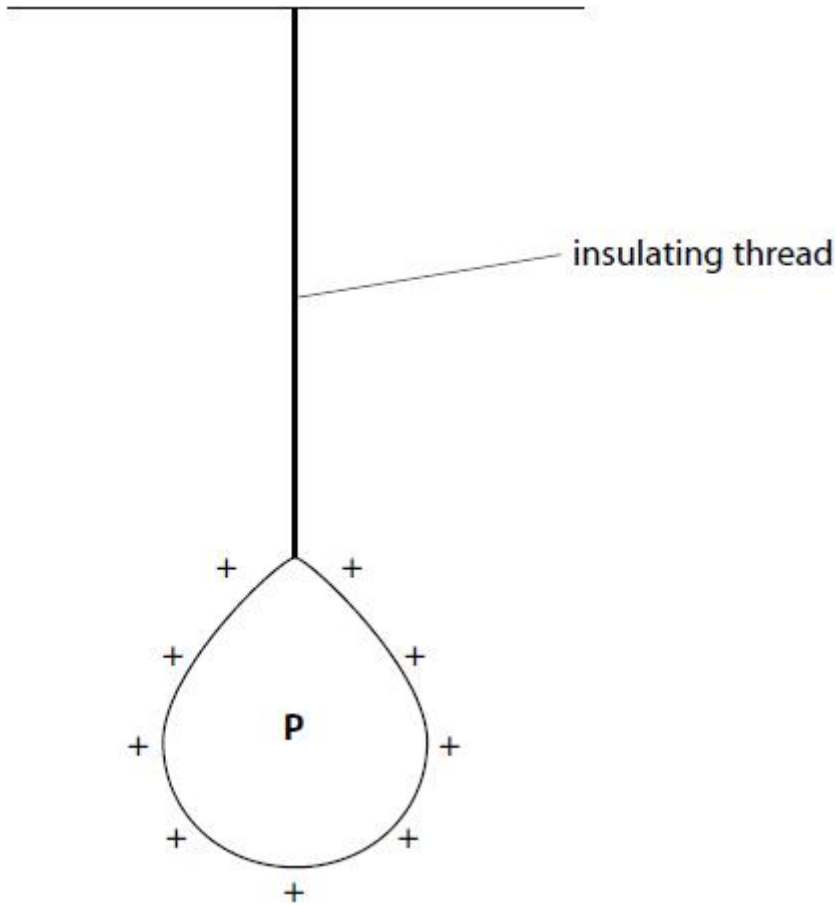


Figure 15

Two more balloons, **Q** and **R**, are charged and placed either side of balloon **P**.

The balloons move to the positions shown in Figure 16.

Add the charges on balloons **Q** and **R** in Figure 16.

(2)

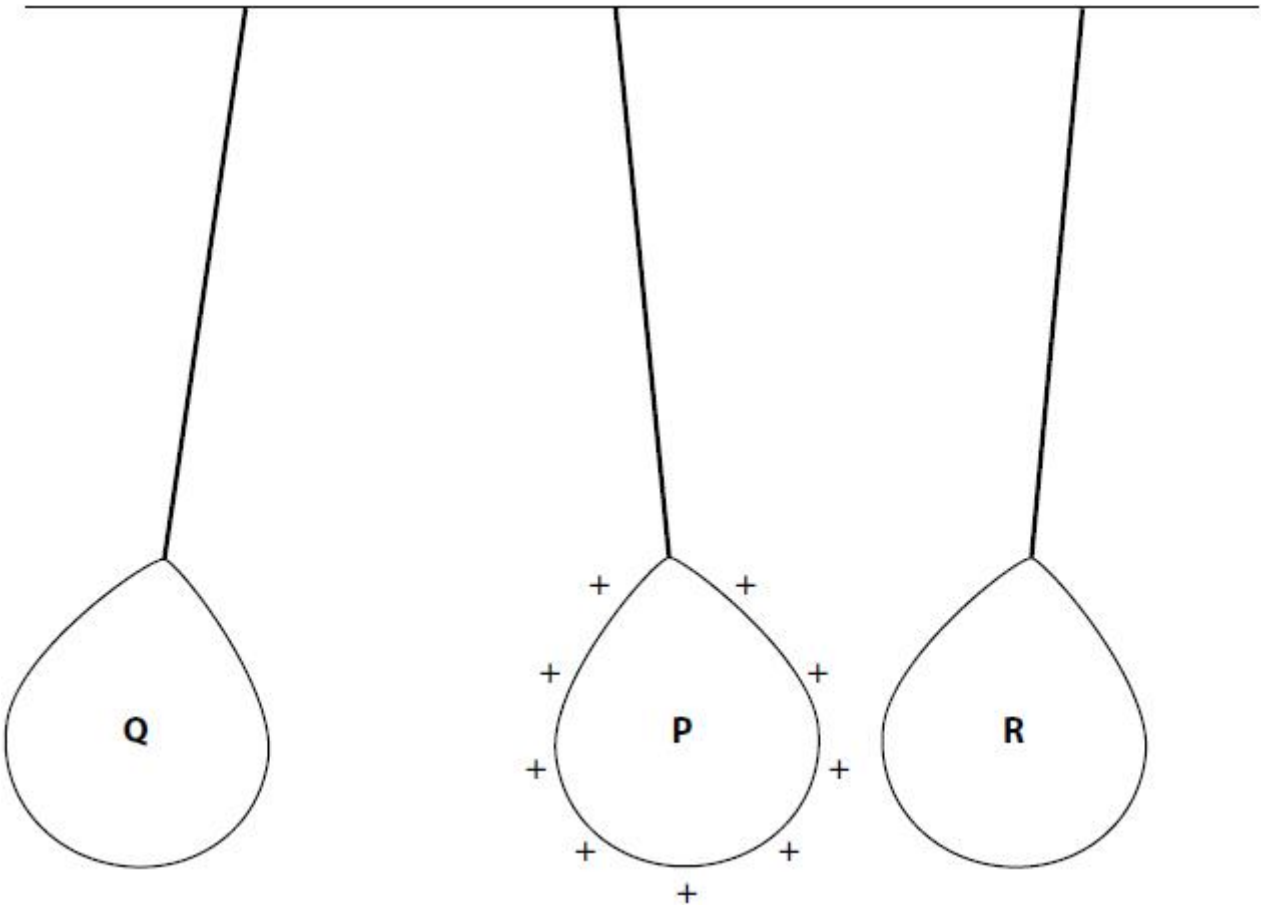
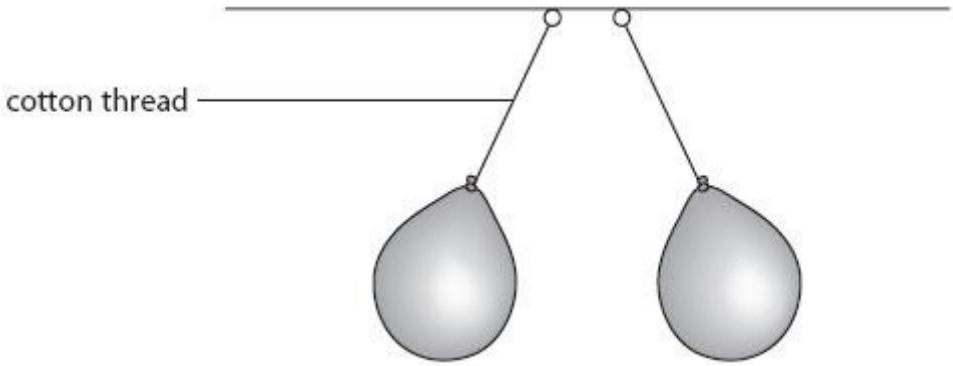


Figure 16

(Total for question = 2 marks)

Q2.

A student charges two balloons and hangs them side by side.



Explain why the cotton threads are not vertical.

(2)

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Q3.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 20 shows a device that removes small particles of soot from sooty air.

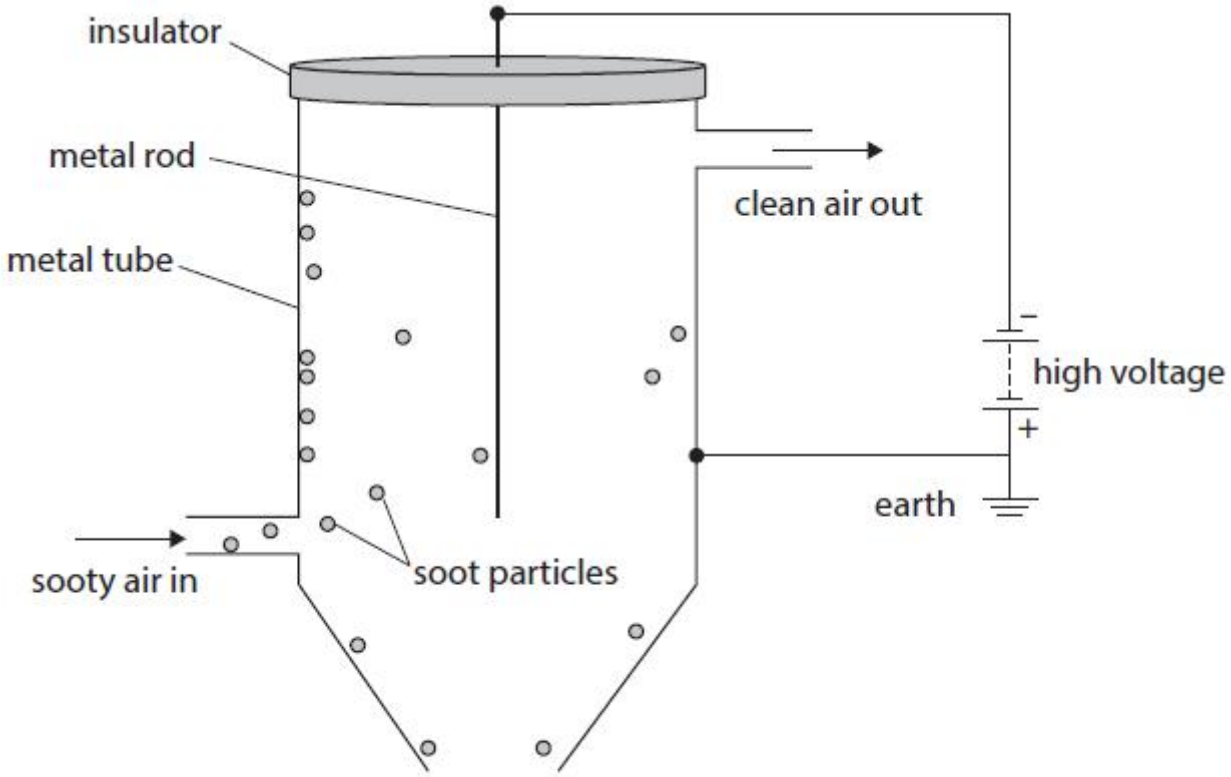

Figure 20

(i) There is a large potential difference between the metal tube and the metal rod in the centre of the tube.

This large potential difference causes the soot particles to gain a negative charge as they pass between the rod and the tube.
The soot particles have

(1)

☐ **A** lost protons
☐ **B** lost electrons
☐ **C** gained protons
☐ **D** gained electrons

(ii) Explain why, in Figure 20, the charged soot particles move towards the metal tube.

(2)

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(iii) When the soot particles reach the metal tube, the charge on them changes.
State how the charge on them changes.

(1)

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(iv) The metal tube does not become charged when the soot particles reach it.
Explain why the metal tube does not become charged.

(2)

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(Total for question = 6 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about static electricity.

A student has a rubber balloon tied to a long piece of cotton thread.

The student gives the balloon an overall electrostatic charge.

(i) Describe **one** way that the student could give the balloon an overall electrostatic charge.

(2)

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.....

(ii) The student gives the balloon an overall negative charge.
Which of these sentences explains why the overall charge on the balloon is negative?

(1)

- ☐ **A** Negative charge has been removed from the balloon.
- ☐ **B** Negative charge has been added to the balloon.
- ☐ **C** Positive charge has been removed from the balloon.
- ☐ **D** Positive charge has been added to the balloon.

(iii) The student charges another balloon on a long thread.

Explain how the student can show that the two balloons have the same type of charge.

(3)

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(Total for question = 6 marks)

Q5.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A cleaning company uses an electric sprayer to produce a cloud of disinfectant inside a bus.

The sprayer gives a negative electric charge to each droplet in the cloud of disinfectant.

(i) The droplets in the cloud of disinfectant have a negative charge because the sprayer

(1)

- ☐ **A** adds electrons to the droplets
- ☐ **B** adds protons to the droplets
- ☐ **C** removes electrons from the droplets
- ☐ **D** removes protons from the droplets

(ii) Figure 16 shows the shape of the clouds of droplets in the disinfectant from two sprayers.

Figure 16

Sprayer 1 gives a negative charge to each droplet.

Sprayer 2 does not give any charge to the droplets.

Explain the difference in the shape of the two clouds.

(2)

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(iii) Figure 17 shows a cloud of disinfectant, containing charged droplets, moving towards a seat in the bus.

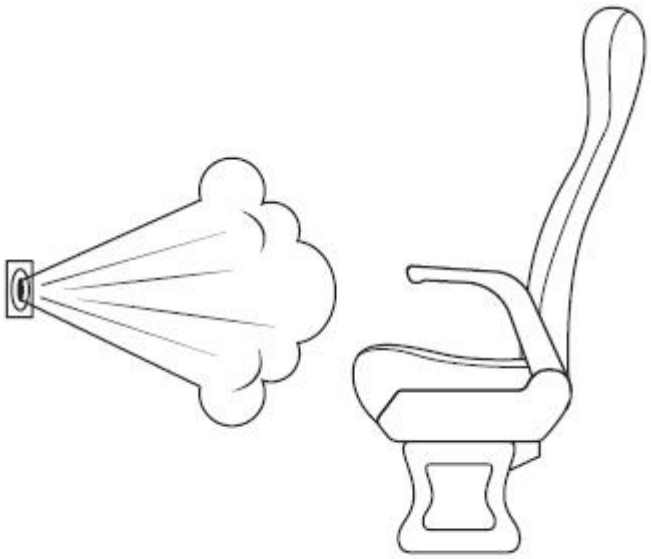

Figure 17

Explain how charging the droplets in the cloud of disinfectant makes sure that every part of the seat is disinfected.

(2)

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.....

(Total for question = 5 marks)

Q6.

Complete the following sentences using words from the box below.

electrostatic	friction	gravitational	magnetic

(i) The force that keeps the Earth in orbit around the sun is
..... attraction.

(1)

(ii) The force that opposes motion between two surfaces is
..... .

(1)

(iii) The force that can move small pieces of paper towards a plastic comb
is ..... attraction.

(1)

(Total for question = 3 marks)

Q7.

A student rubs **two** balloons with a dry cloth.

The balloons become positively charged.

The student hangs the charged balloons using strings, as shown in Figure 5.

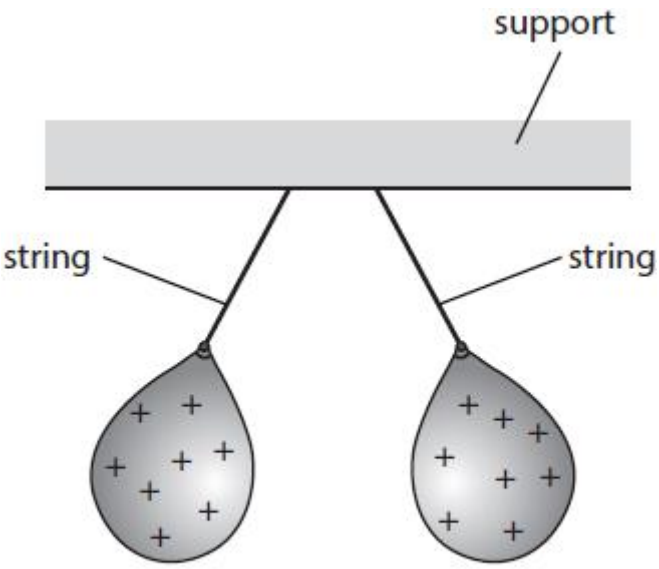

Figure 5

Use words from the box to complete the sentences.

(4)

ampere	attract	coulomb	electrons
negative	positive	protons	repel

The balloons have the same charge.

This means that these balloons ..... each other.

The charged particles transferred from the balloons to the cloth are called ..... .

The cloth is left with a ..... charge.

The unit of charge is the ..... .

(Total for question = 4 marks)

Q8.

Describe **one** situation where separation of electric charge can create a spark.

(2)

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.....

Q9.

This question is about static electricity.

Draw **one** straight line from each example of electrostatic charges in action to their descriptions.

(3)

electrostatic charges in action	description
charging a plastic comb	small droplets are charged so they will stick to an object
electrostatic paint spraying	build-up of charge in a cloud causes a discharge to Earth
safe fuelling of cars by earthing	prevents a dangerous build-up of charge between a flowing liquid and a pipe
lightning	produced by friction between solid surfaces

(Total for question = 3 marks)

Q10.

Figure 15 shows two parallel metal plates.

The plates are charged using a very high voltage.

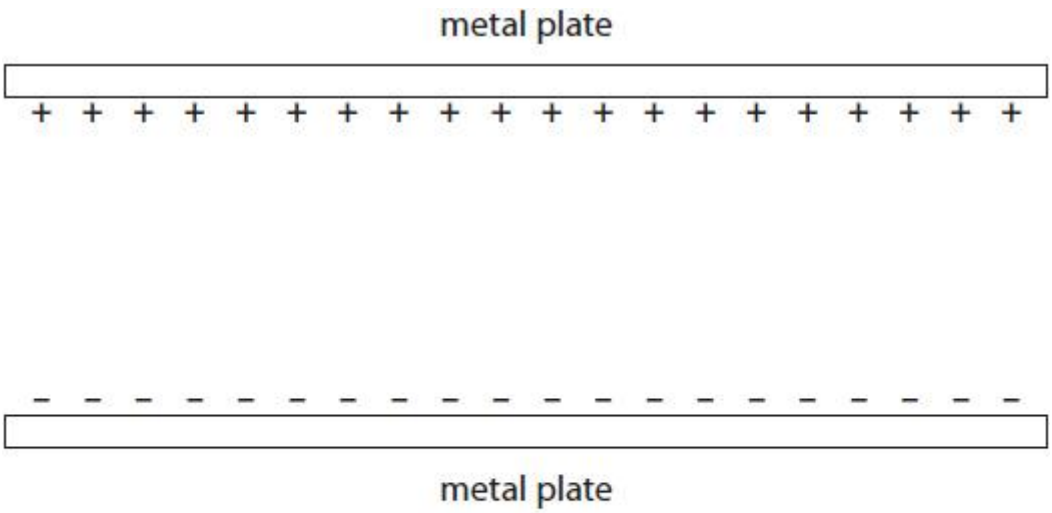


Figure 15

Draw on Figure 15 the shape and direction of the electric field **between** the plates.

(2)

(Total for question = 2 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	• label to indicate that balloon Q has a positive charge (1) • label to indicate that balloon R has a negative charge (1)	accept responses showing appropriate +/− signs or worded label	(2)

Q2.

	Answer	Acceptable answers	Mark
	an explanation linking: balloons repel (1) (because) they have like charges (1)	balloons repulse / push away (from each other/to the side) same charge / both positive / both negative accept like charges repel for 2 marks	**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
(i)	D gained electrons A and C are incorrect because protons are not exchanged in this context B is incorrect because it would give the particles a net positive charge		**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking any **two** from (soot particles are in an) electric field (1) **AND EITHER** attraction (1) (surface of) it/tube has (induced) positive/opposite/unlike charge (1) **OR** repulsion (1) rod (and soot particles) has a negative/like/same charge (1)		**(2)** **AO1**

Question number	Answer	Additional guidance	Mark
(iii)	(net charge) becomes zero (1)	discharged / charge is lost / charge moves on to the metal tube/ becomes neutral allow electrons for charge ignore becomes positively charged	**(1)** **(AO1)**

Question number	Answer	Additional guidance	Mark
(iv)	an explanation linking metal is a conductor (1) (connected to) earth/ground (1)	allow electrons for charge charge moves through the metal / tube / wire	**(2)** **AO1**

Q4.

Question number	Answer	Additional guidance	Mark
(i)	rub (the balloon) (1) with a piece of cloth/hair/fur (1)	use friction allow on any insulated object	**(2)**

Question number	Answer	Additional guidance	Mark
(ii)	B Negative charge has been added to the balloon A is incorrect removing negative charge would make the balloon positively charged. C and D are incorrect because positive charge cannot be moved		**(1)**

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking: method of handling balloons without discharging them (1) bring balloons near to each other (1) observation of repulsion (1)	 hang balloons up by their strings they/balloons will push away (from each other)	(3)

Q5.

Question number	Answer	Additional guidance	Mark
(i)	A adds electrons to the droplets B and D are incorrect because protons cannot be added or removed from droplets. C is incorrect because this would give an overall positive charge		(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking cloud from sprayer 1 is more dispersed (than from sprayer 2) (1) the droplets / charges repel each other (1)	accept reverse argument spread out more (covers) larger area bigger / wider	(2) AO3

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking (droplets) attracted (to seat) (1) including parts not in direct line of spray (1)	 induced charge (on seat) spreads out (over all the seat)	(2) AO2

Q6.

Question number	Answer	Mark
	(i) gravitational (1) (ii) friction (1) (iii) electrostatic (1)	(3) AO1

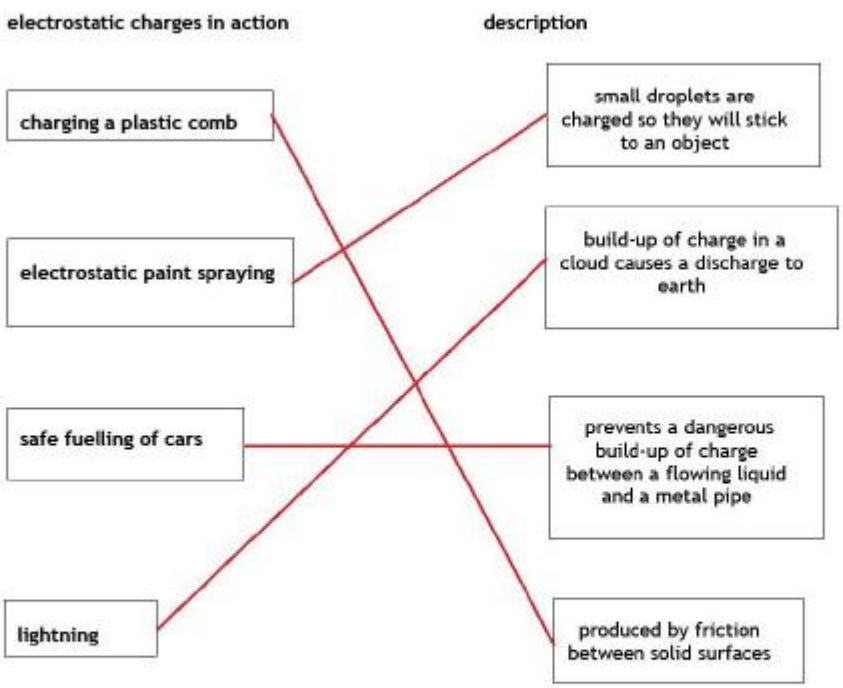
Q7.

Question	Answer	Additional guidance	Mark
	The balloons have the same charge. This means that these balloons repel each other. (1) The charged particles transferred from the balloons to the cloth are called electrons (1) The cloth is left with a negative charge. (1) The unit of charge is the coulomb (1)	more than one word in a space loses the mark for that space	(4) AO1.1

Q8.

	Answer	Acceptable answers	Mark
	A description to include the situation which caused the charge separation (1) where the spark travelled from or to(1)	examples when refuelling, spark between end of fuel/pipe and vehicle =2 spark between/from /to person comb/clothes/metal handle and, when combing hair/removing clothing/opening door = 2 lightning flash, between cloud and cloud/plane/ground, =2 ignore between plug and socket/jump leads	**(2)**

Q9.

Question number	Answer	Additional guidance	Mark
		1 mark for one line correct 2 marks for two or three lines correct 3 marks for four lines correct. if more than one line is drawn from or to a box, do not credit any of those lines	**(3)** **AO3**

Q10.

Question number	Answer	Additional guidance	Mark
	drawing lines to include any **two** from		**(2) AO1**
	any vertical line (in between plates) (1)	judge by eye	
	at least two parallel lines (1)	ignore any curved lines at the ends of the plates	
	any arrow downwards (to show direction) (1)	reject contradicting arrows for this mp	